RM520N Series

Reference Design

5G Module Series

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Status: Released

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About the Document

Revision History

Version	Date	Author	Description
-	2021-12-13	Wynna SHU	Creation of the document
1.0	2022-07-15	Wynna SHU	First official release
1.1	2023-04-17	Archibald JIANG	1. Added applicable module: RM520N-EU. 2. Updated the value of R0202, R0204, R0205, R0206, R0207, R0208, C0212, L0201, Vstart, Vstop and Vref (Sheet 2).
1.2	2023-12-22	Archibald JIANG	Added a note on avoiding abnormal RF functions caused by current sink on the module's pins (Sheet 1).

Contents

About the Document **3**
Contents **4**

1 Reference Design **5**
1.1. Introduction 5
1.2. Schematics 5

1 Reference Design

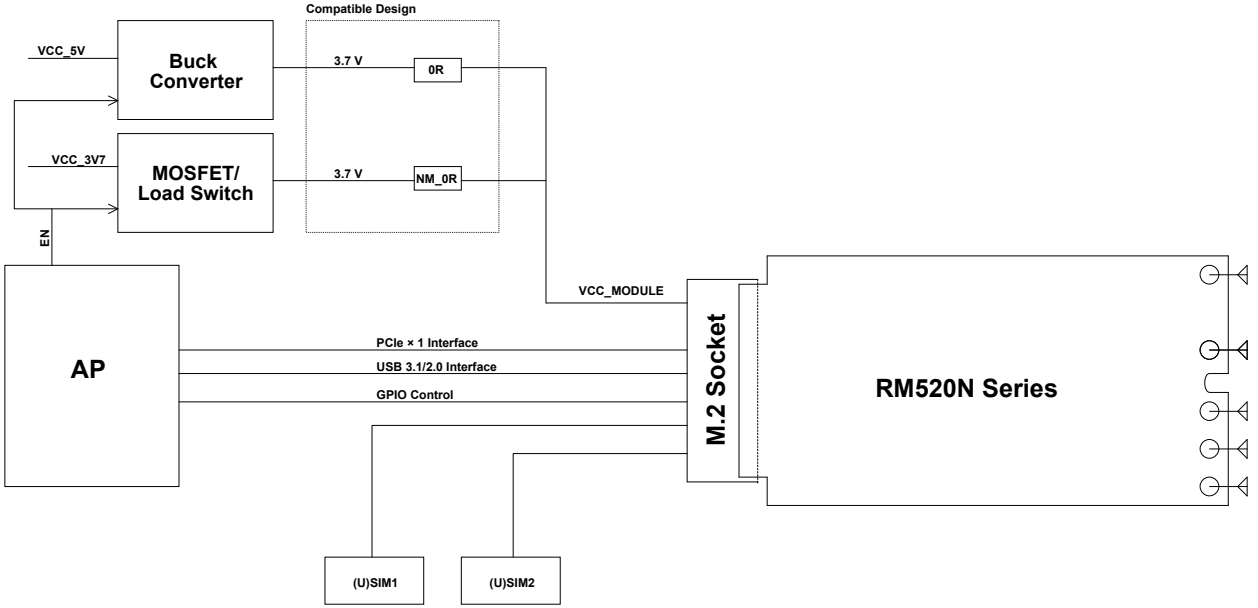
1.1. Introduction

This document provides the reference design for RM520N series module, including the design for power supply, module interfaces, AP interfaces and (U)SIM interfaces.

1.2. Schematics

The schematics illustrated in the following pages are provided for reference only.

Block Diagram

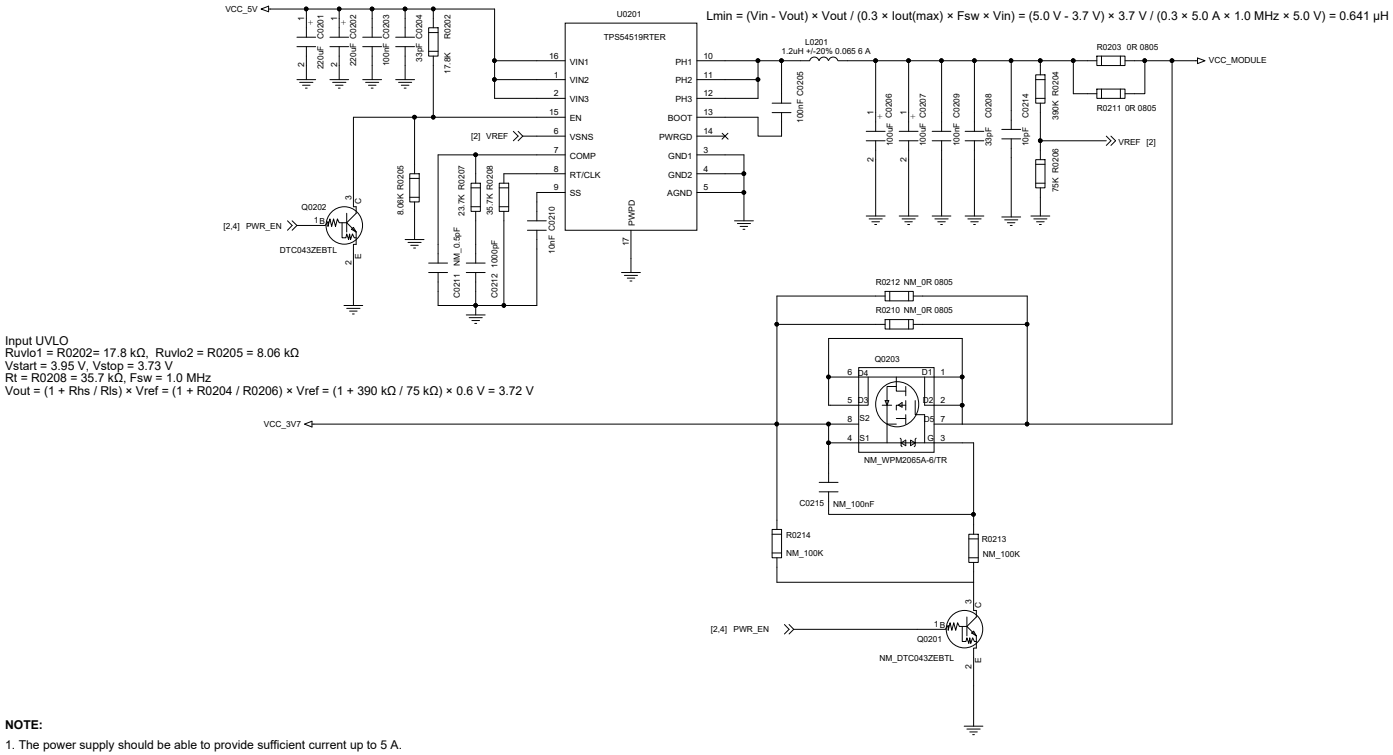


NOTE:

1. The location of anntena connectors is for reference only.
2. RM520N-EU supports 5 antennas (ANT0, ANT1, ANT2, ANT3 and ANT4), while RM520N-GL only supports 4 antennas (ANT0, ANT1, ANT2 and ANT3).
3. Before the module turns on, ensure the pins DPR and USIM_DET are not pulled high to avoid current sink damaging the module. For more details, contact Quectel Technical Support.

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Power Supply Design



Lmin = (Vin - Vout) × Vout / (0.3 × Iout(max) × Fsw × Vin) = (5.0 V - 3.7 V) × 3.7 V / (0.3 × 5.0 A × 1.0 MHz × 5.0 V) = 0.641 μH

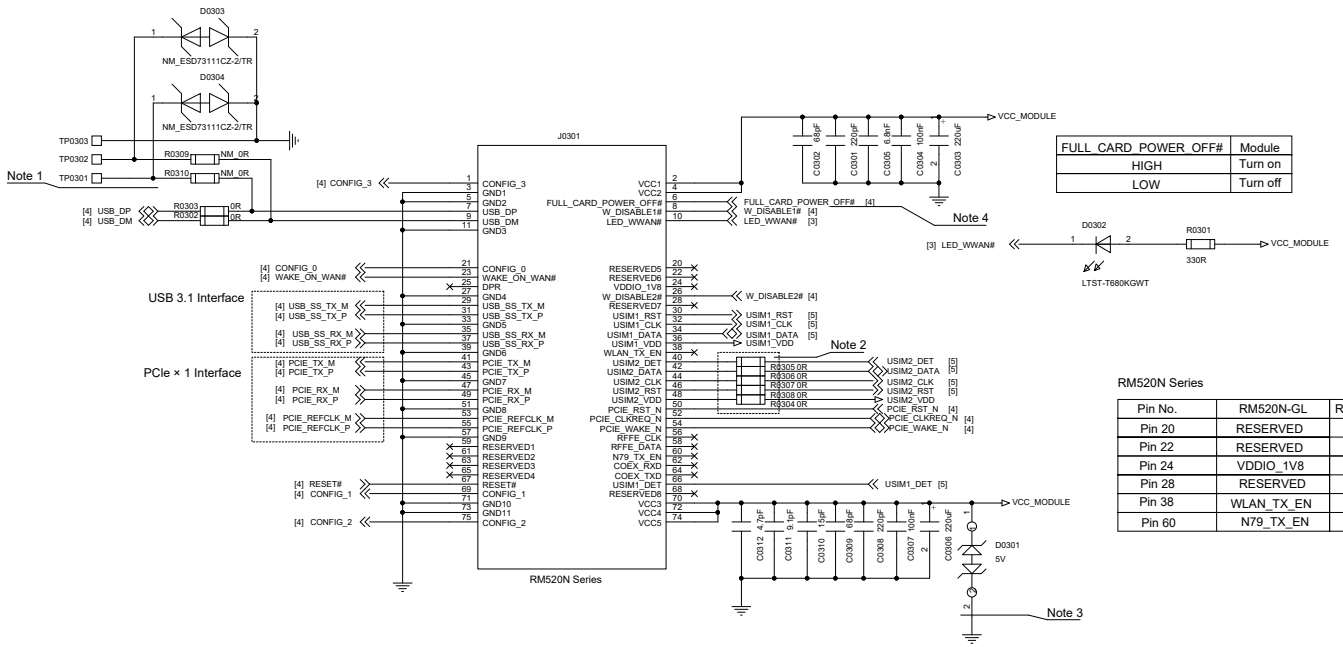
Input UVLO
Ruvlo1 = R0202= 17.8 kΩ, Ruvlo2 = R0205 = 8.06 kΩ
Vstart = 3.95 V, Vstop = 3.73 V
Rt = R0208 = 35.7 kΩ, Fsw = 1.0 MHz
Vout = (1 + Rhs / Rls) × Vref = (1 + R0204 / R0206) × Vref = (1 + 390 kΩ / 75 kΩ) × 0.6 V = 3.72 V

NOTE:

1. The power supply should be able to provide sufficient current up to 5 A.
2. A compatible power supply design for the module is recommended.

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DATE: Friday, December 22, 2023	SHEET: 2 OF 5	

Module Interfaces



FULL_CARD_POWER_OFF#	Module
HIGH	Turn on
LOW	Turn off

RM520N Series

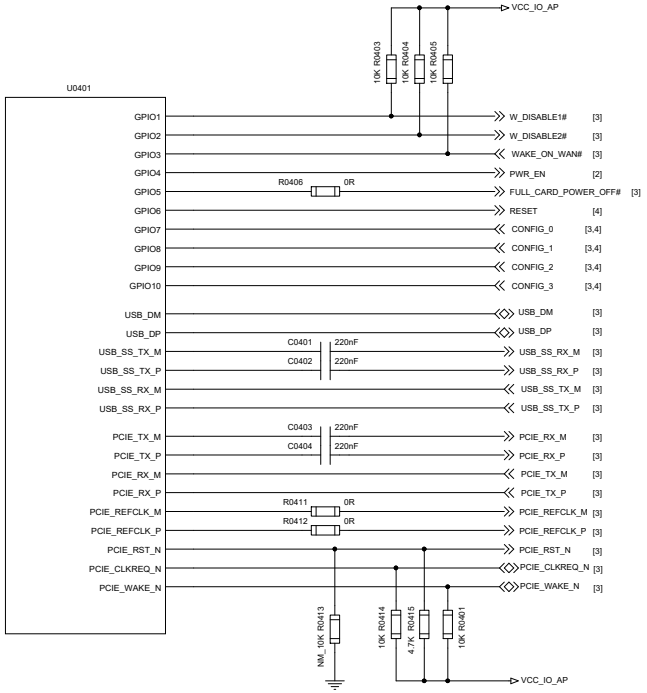
Pin No.	RM520N-GL	RM520N-EU
Pin 20	RESERVED	PCM_CLK
Pin 22	RESERVED	PCM_DIN
Pin 24	VDDIO_1V8	PCM_DOUT
Pin 28	RESERVED	PCM_SYNC
Pin 38	WLAN_TX_EN	NC
Pin 60	N79_TX_EN	NC

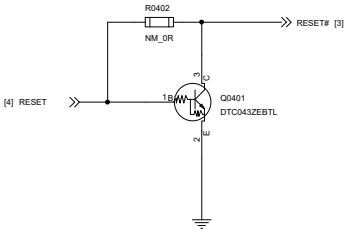
NOTE:

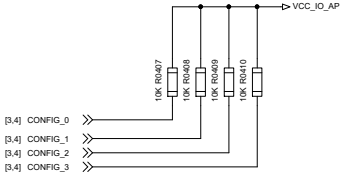
1. Test points must be reserved for the firmware upgrade over USB interface and to minimize the stub length of USB test signals.
2. R0304 to R0308 should be placed close to the M.2 socket. If the module has a built-in eSIM, R0304 to R0308 should not be mounted.
3. It is recommended to use a TVS with working peak reverse voltage of 5 V and it should be placed close to the M.2 socket.
4. Use an AP GPIO to control FULL_CARD_POWER_OFF# of the module.

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DATE: Friday, December 22, 2023	SHEET: 3 OF 5	

AP Interfaces







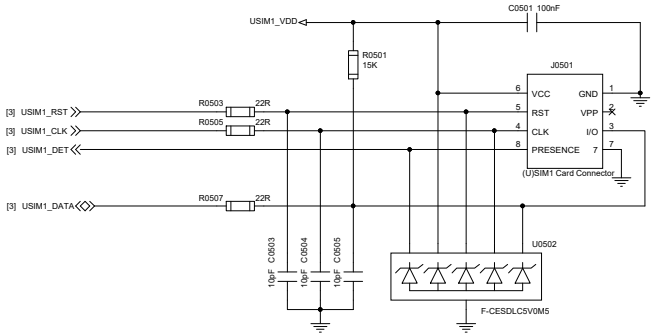
NOTE:

1. U0401 represents your host.
2. Keep C0401–C0404 to the host as close as possible.
3. The differential impedance of USB 2.0 and USB 3.1 signal traces should be controlled to 90 Ω.
4. The differential impedance of PCIe signal traces should be controlled to 85 Ω.
5. If a USB connector is used, please keep ESD protection components to the USB connector as close as possible.

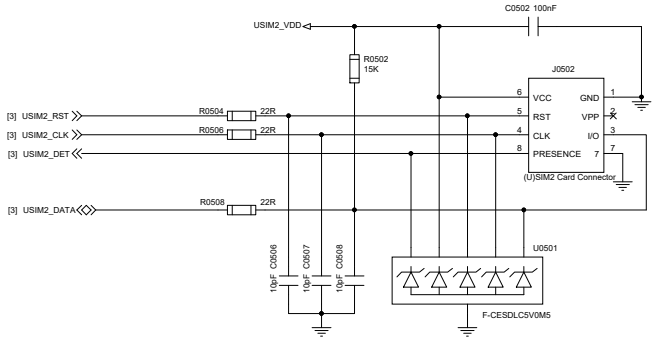
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(U)SIM Interfaces Design

(U)SIM 1



(U)SIM 2



NOTE:

1. The decoupling capacitors of (U)SIM signals and (U)SIM related resistors must be placed close to (U)SIM card connectors.
2. The module provides the input pins USIM1_DET and USIM2_DET to detect (U)SIM cards.
   1) In this reference design, a normally short-circuited (U)SIM card connector is used and high-logic-level detection is supported. For more details, see the corresponding Hardware Design document.
   2) USIM1_DET and USIM2_DET are pulled LOW by default, and will be internally pulled up to 1.8 V by software configuration only when (U)SIM hot-plug is enabled by AT+QSIMDET.
3. R0503–R0508 are used to suppress the EMI such as spurious transmission.
4. An ESD diode array with a junction capacitance of no higher than 10 pF should be placed as close to the (U)SIM card connector as possible for ESD protection.
5. The (U)SIM card connector should be placed close to the M.2 socket, because a long PCB trace may lead to waveform distortion, which affects the signal quality.

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DATE: Friday, December 22, 2023		SHEET: 5 OF 5	